

Computational Thinking

## Discrete Mathematics

Number Theory

Topic 00 : Module Introduction

Logic

Lecture 01 : Module Overview

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Graphs and  
Networks

Autumn Semester, 2026/27

Collections

### Outline

- Motivation and aim of this module.
- Administration trivia — Contact hours, Assessment structure, ...
- Resources

Enumeration

Relations & Functions

# Outline

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|                                                         |    |
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# What?

(Aim)

## Aim, as per the Module Descriptor ...

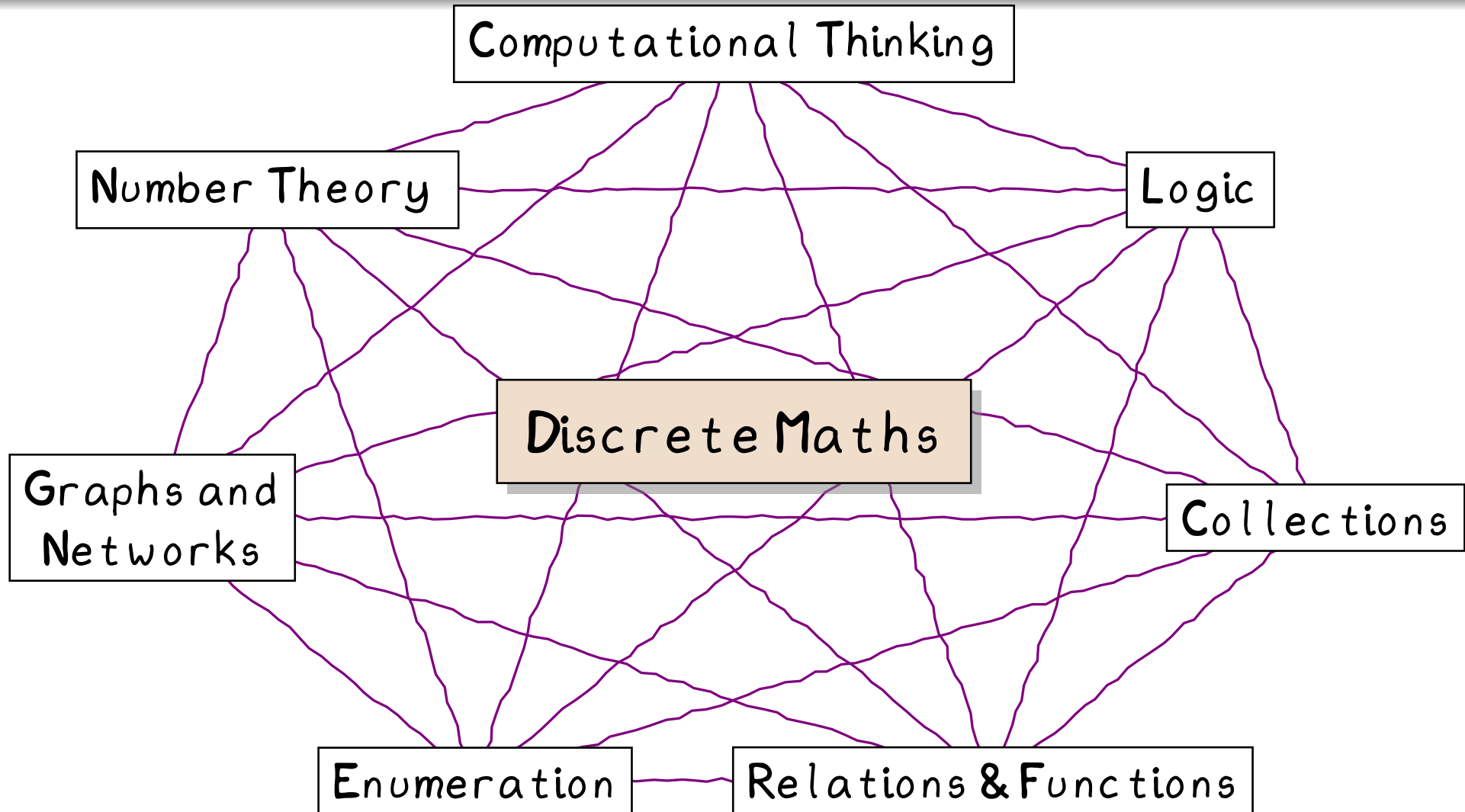
This module provides a solid foundation of selected topics in discrete mathematics related to computing and information sciences. The topics are covered in an elementary manner in order to reinforce understanding of concepts and improving algebraic problem-solving skills so that the student can effectively proceed with their study of a degree programme in computing.

## Translation (Informal Aims)

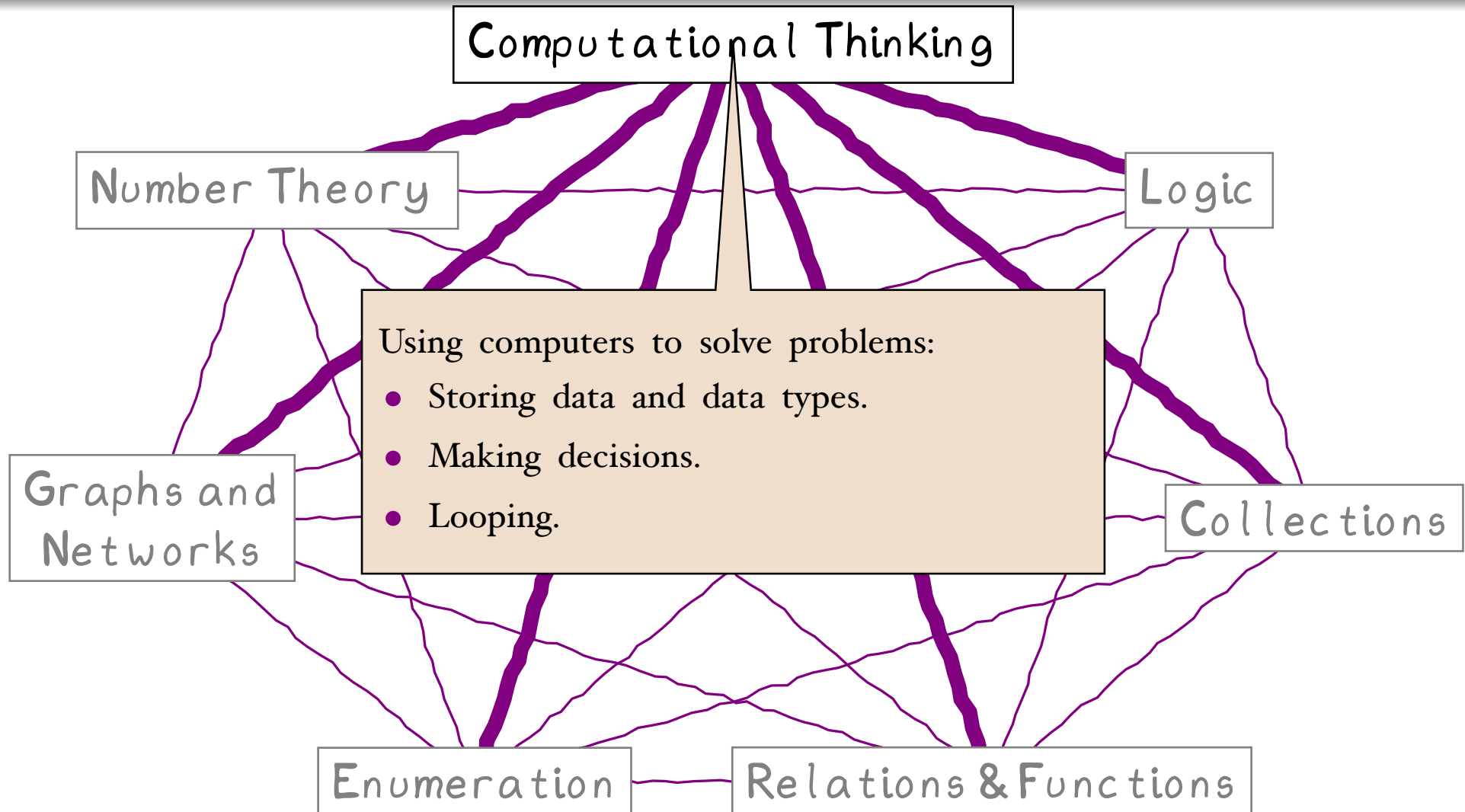
- ① Reason logically — aim for precision and correctness over speed.
- ② Develop and manipulate theoretical models  
— a **set** is a collection of things, a **relation** is a collection of pairs of things, a **graph** is a collection of things with pairwise connections, etc..
- ③ Translate  
computing concepts/implementation (Python)  $\leftrightarrow$  theoretical models (mathematics).

# What?

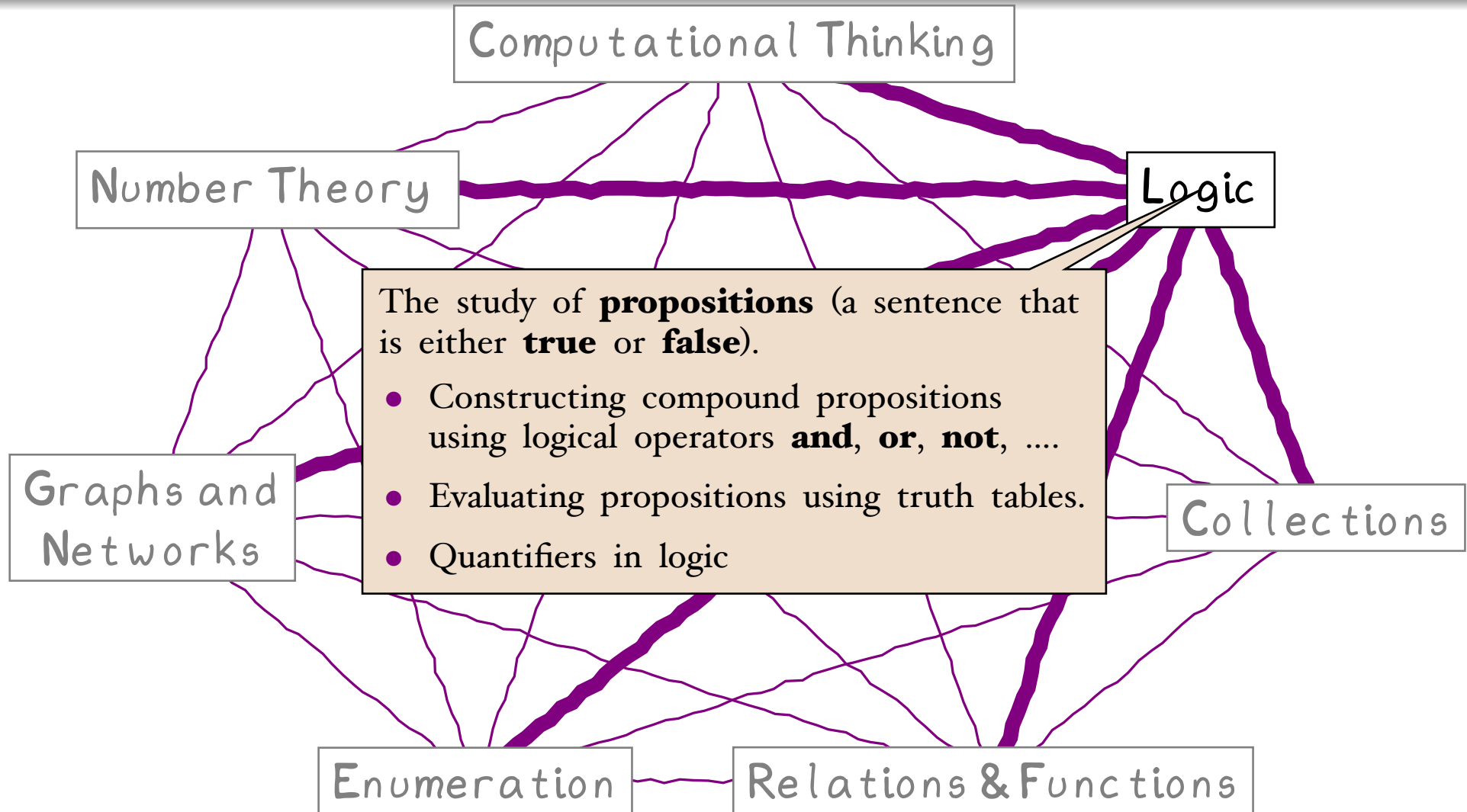
# (Discrete Mathematics)



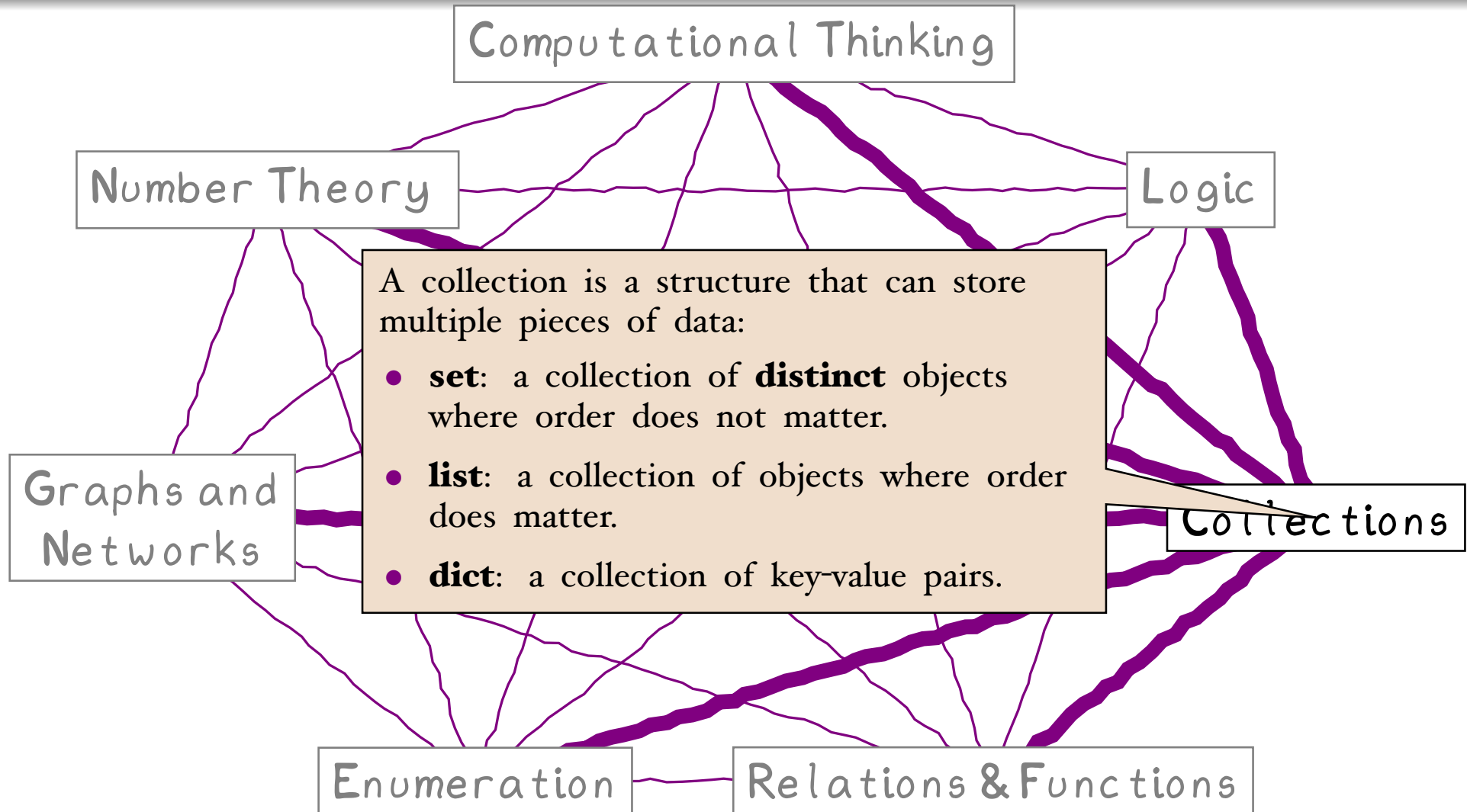
# What?



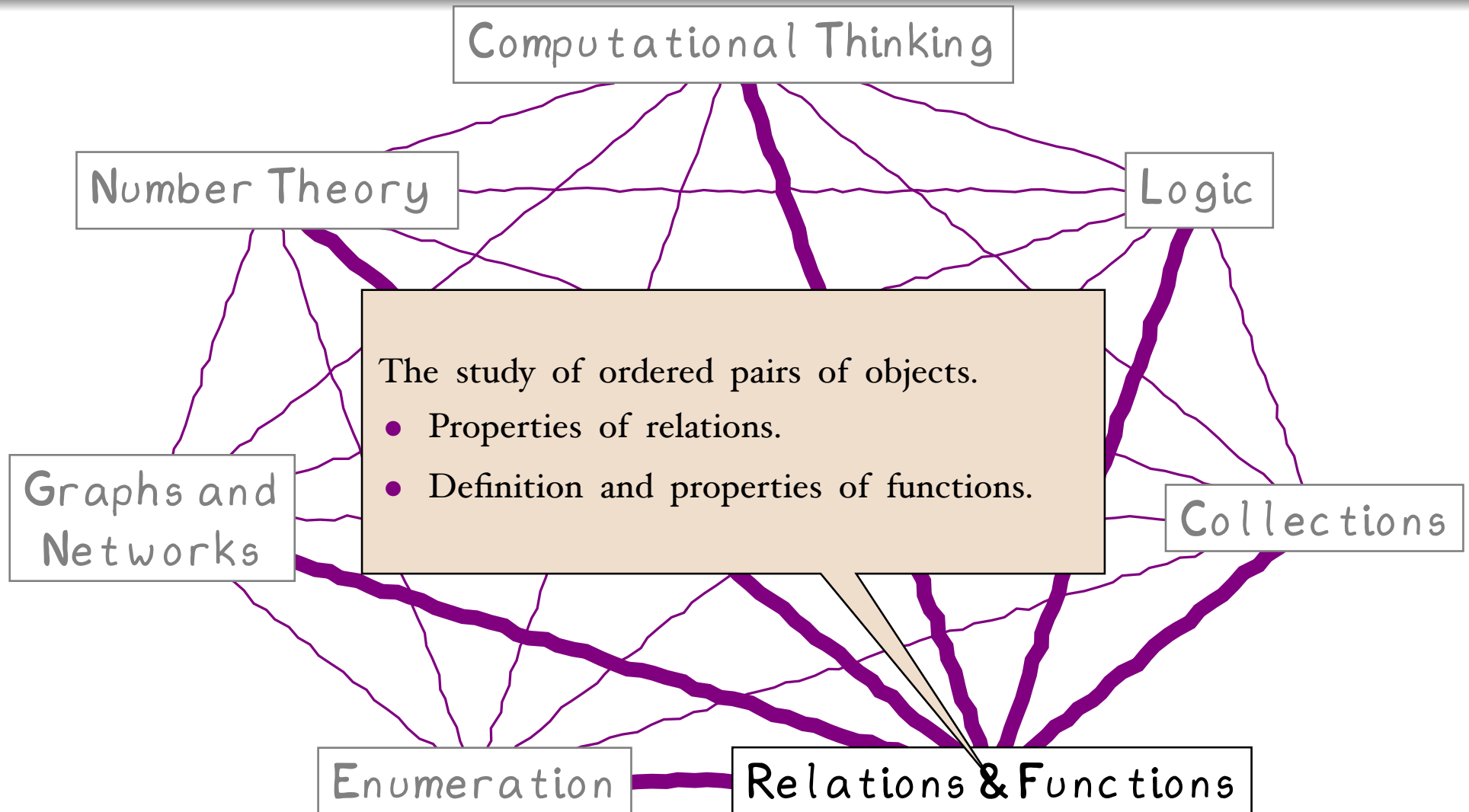
# What?



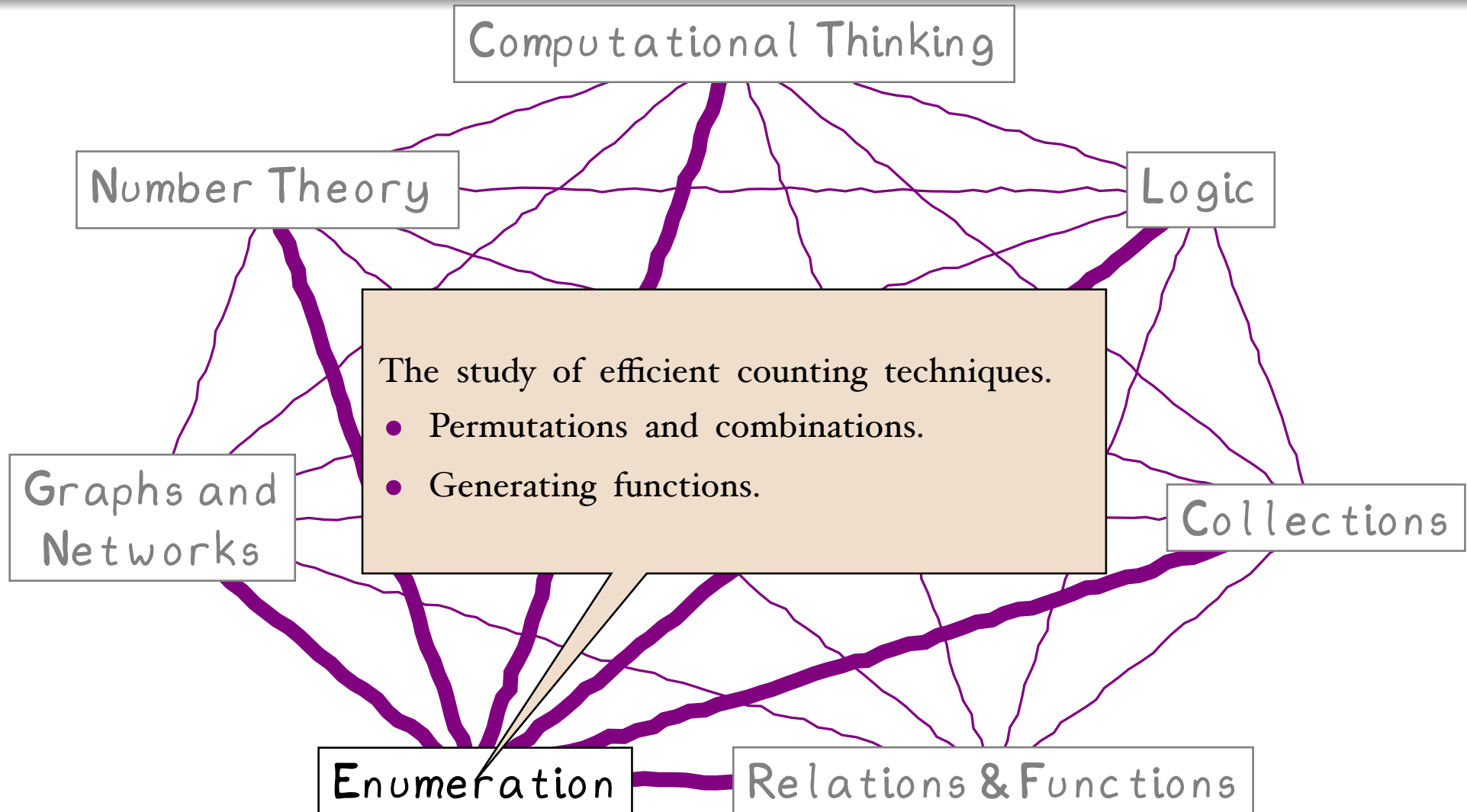
# What?



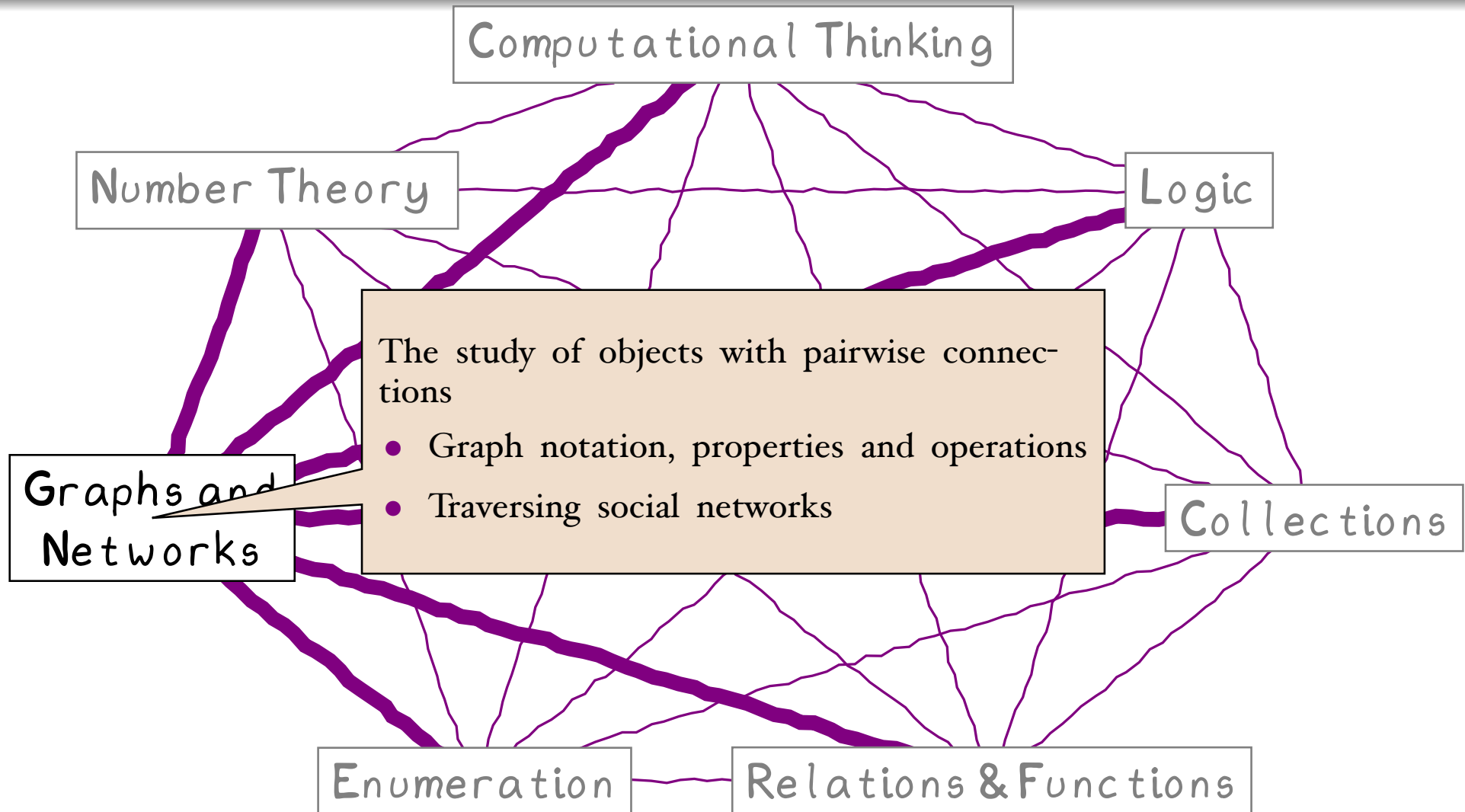
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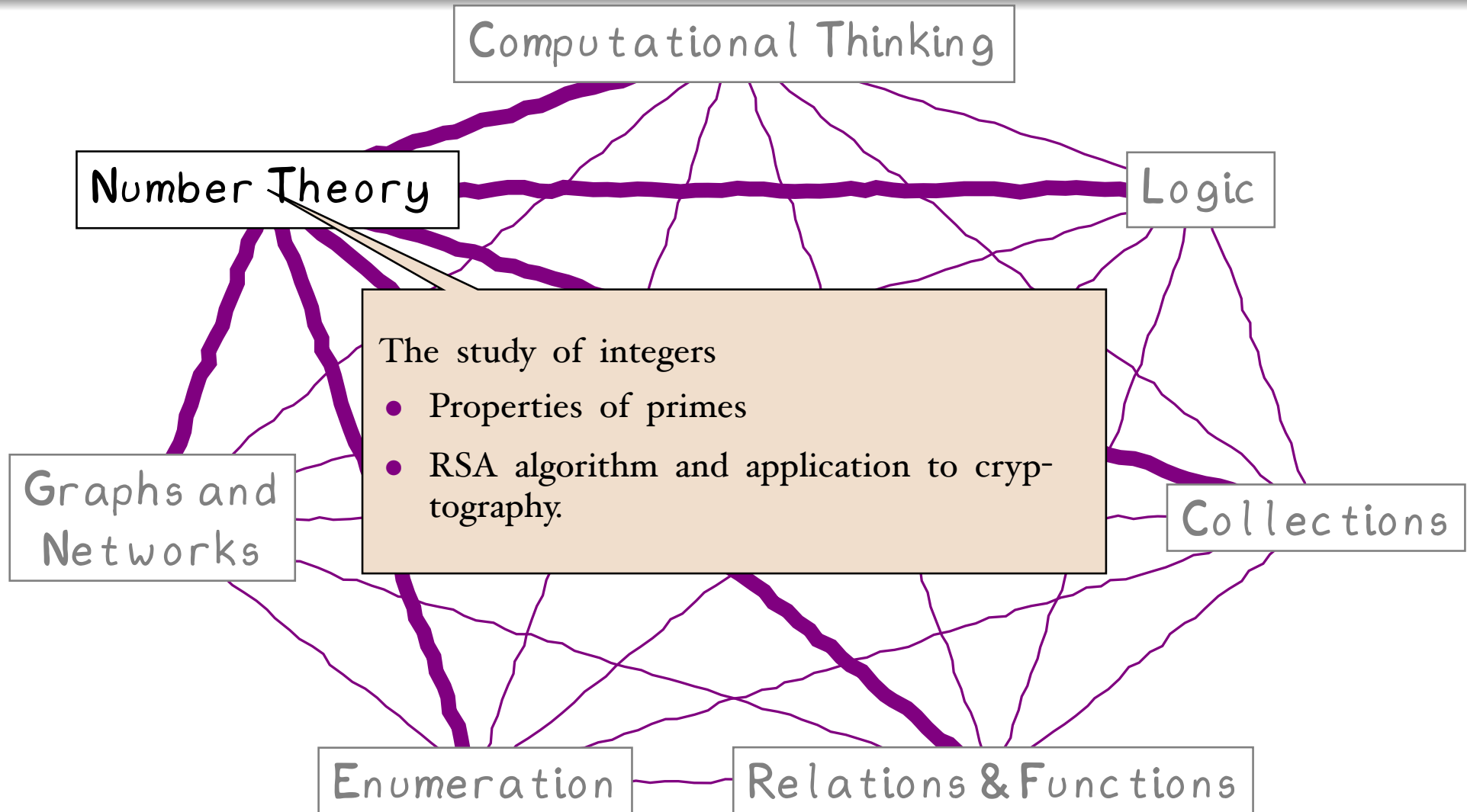
# What?



# What?



# What?

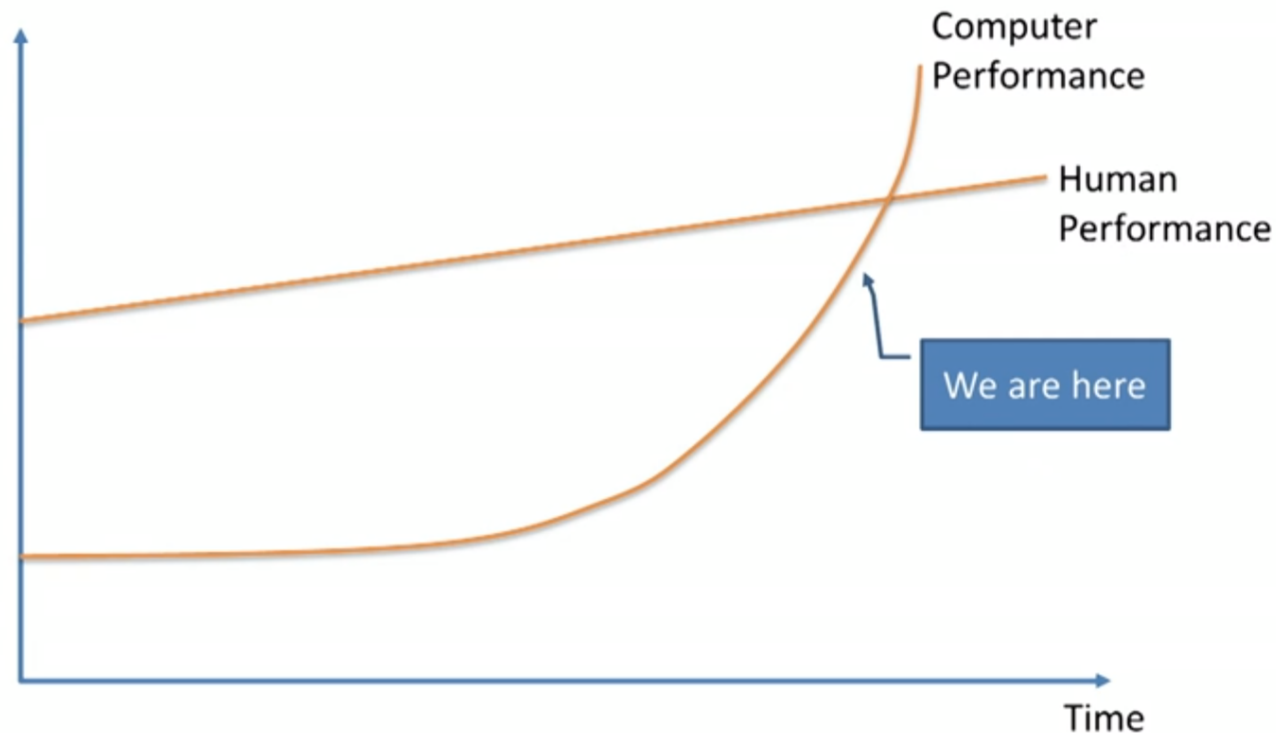


# Why?

Many, many reasons ... picking one\* ...

AI / Machine learning is the future of computing

Discrete mathematics is the core of machine learning.



\*The AI Revolution: Our Immortality or Extinction

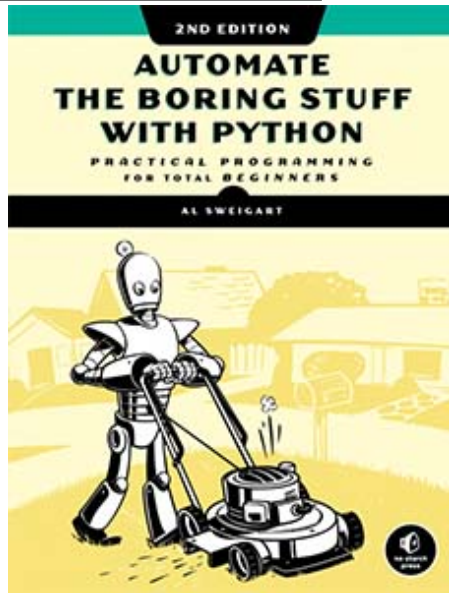
[waitbutwhy.com/2015/01/artificial-intelligence-revolution-2.html](http://waitbutwhy.com/2015/01/artificial-intelligence-revolution-2.html)

# Why?

(Aim)

Computers are faster and more accurate\* than us, they don't get board/distracted†  
⇒ Get computers to do the hard (and/or boring) stuff

See Al Sweigart's books



Or our view on life



Why have a dog and bark?

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\*Exception to this are the generative text models (chatGPT\*, Bard, ...) where accuracy is replaced by "most likely/probable".

†Well, if you don't count Teslas, in FSD mode, driving over pedestrians ...

# Aside: Speaking of Elon ... AI and all that ...

## List of predictions for autonomous Tesla vehicles by Elon Musk

From Wikipedia, the free encyclopedia

| Date ↕          | Prediction ↕ | Quote                                                                                                                                                | Met ↕                                                |
|-----------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Sep 2013</b> | 2016         | "We should be able to do 90 percent of miles driven [autonomously] within three years."                                                              | ✗ No                                                 |
| <b>Jun 2014</b> | 2015         | "I am confident that in less than a year you will be able to go from highway on ramp to highway exits without touching any controls."                | ✗ No                                                 |
| <b>Oct 2014</b> | 2015         | "A Tesla car next year will probably be 90-percent capable of autopilot. Like, so 90 percent of your miles can be on auto. For sure highway travel." | AP/HW1 released Oct 2015 for highways <sup>[7]</sup> |
| <b>Oct 2014</b> | 2023         | "I think we'll be able to achieve true autonomous driving, where you could literally get in the car, go to sleep and wake up at your destination,"   | ✗ No                                                 |
| <b>Oct 2015</b> | 2018         | "From a technology standpoint, Tesla will have a car that can do full autonomy in about three years, maybe a bit sooner."                            | ✗ No                                                 |

|                 |      |                                                                                                                                                                                                                                                                                                   |      |
|-----------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Dec 2015</b> | 2018 | "We're going to end up with complete autonomy, and I think we will have complete autonomy in approximately two years."                                                                                                                                                                            | ✗ No |
| <b>Jan 2016</b> | 2018 | "Ultimately you'll be able to summon your car anywhere ... your car can get to you. I think that within two years, you'll be able to summon your car from across the country. It will meet you wherever your phone is ... and it will just automatically charge itself along the entire journey." | ✗ No |
| <b>Jun 2016</b> | 2019 | "I consider autonomous driving to be a basically solved problem. ... We're less than two years away from complete autonomy. Regulators however will take at least another year; they'll want to see billions of miles of                                                                          | ✗ No |

# How?

(Contact Hours)

## ⚙️ Three lectures per week

- ⚙️ Cover concepts, definitions, examples, etc.
- ⚙️ BUT feel free to stop me and ask questions at any point.
- ⚠️ **You should print out notes in advance of lecture, or have access to them during class.**
- ⚙️ Ideally you skimmed over the notes in advance of lecture.
- ⚠️ **Take notes during lectures.**

## ⚙️ One tutorial per week

- ⚙️ Review of exercises based on the material covered in the lectures.
- ⚠️ **You need to have printout of tutorial sheets in advance of lecture.**
- ⚙️ Ideally you have attempted/completed some/all questions in advance of tutorial and you are just attending the tutorials to show off.
- ⚙️ Online quiz for self review at end of each topic.

## ⚙️ One practical per week

- ⚙️ Using Python (via *colab notebooks*) to demonstrate implementation details of discrete mathematics concepts.
- ⚙️ Introduce programming in Python — never have too much programming.
- ⚠️ **You need to upload notebook by end of week (Saturday 11:00pm).**

# How?

## (Assessment Structure)

### 75% End of Semester Exam

Current plan (this is subject to change so ask about this in week 10!)

- 4 questions (typically 3–5 parts per question. Answer all questions (i.e. no choice).
- Tend not to have question per topic.
- 80% same material as last year — but there may still be some differences in format/style of questions as will the relative emphasis/weighting of the different topics.

### 25% Continuous Assessment

- Practical work based on 10 python practicals and 2+ online class tests<sup>‡</sup>.
- In theory<sup>§</sup>, weekly assignments, are graded in advance of next week.

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<sup>§</sup>In practice, I ~~may~~ will fall behind a bit.

# Who?

## Dr Denis Flynn

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✉ Denis.Flynn@setu.ie



### Background

- PhD in Applied Mathematics and Civil Engineering.
- BSc (H) Physics.

### Academic Interests

- Dynamical systems, in particular systems with hysteresis
- Game development
- Languages: C/C++, Python, Java

## Dr Kieran Murphy

🏠 Room 313  
☎ 051-30 2055  
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### Background

- PhD in Applied Mathematical Sciences
- BSc (H) Applied Mathematics.

### Academic Interests

- Dynamical systems, in particular numerical analysis
- Game development
- Languages: C/C++, Python, Java

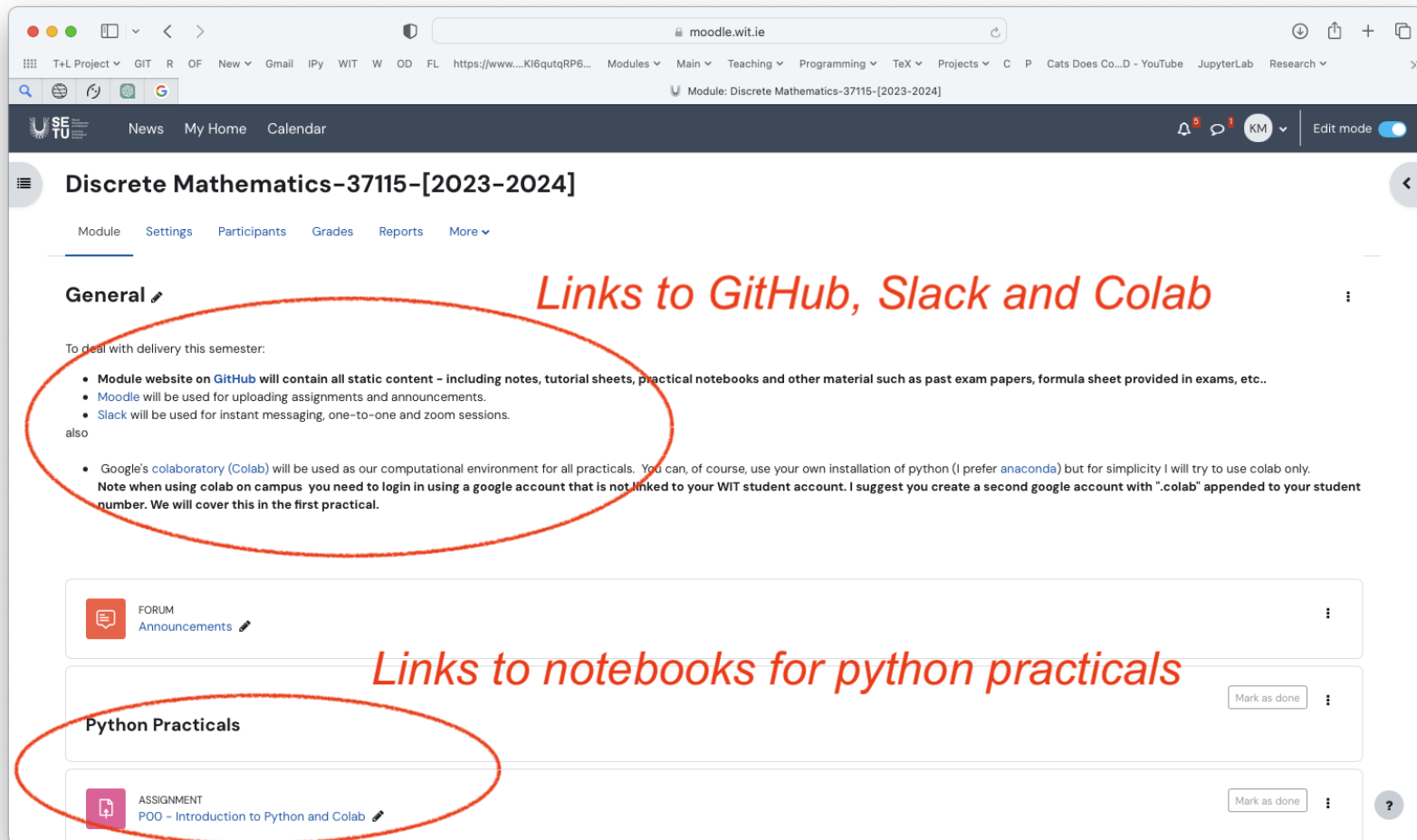
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# Resources — Moodle

- URL: [moodle.wit.ie/course/view.php?id=226896](https://moodle.wit.ie/course/view.php?id=226896)
- Used for all notices, assignment and practical work submissions.



The screenshot displays the Moodle course page for "Discrete Mathematics-37115-[2023-2024]". The page is titled "General" and contains the following content:

- General**
  - To deal with delivery this semester:
  - Module website on [GitHub](#) will contain all static content – including notes, tutorial sheets, practical notebooks and other material such as past exam papers, formula sheet provided in exams, etc..
  - Moodle will be used for uploading assignments and announcements.
  - Slack will be used for instant messaging, one-to-one and zoom sessions.
- also
- Google's [colaboratory \(Colab\)](#) will be used as our computational environment for all practicals. You can, of course, use your own installation of python (I prefer [anaconda](#)) but for simplicity I will try to use colab only.
- Note when using colab on campus you need to login in using a google account that is not linked to your WIT student account. I suggest you create a second google account with ".colab" appended to your student number. We will cover this in the first practical.

Below the general section, there are two sections:

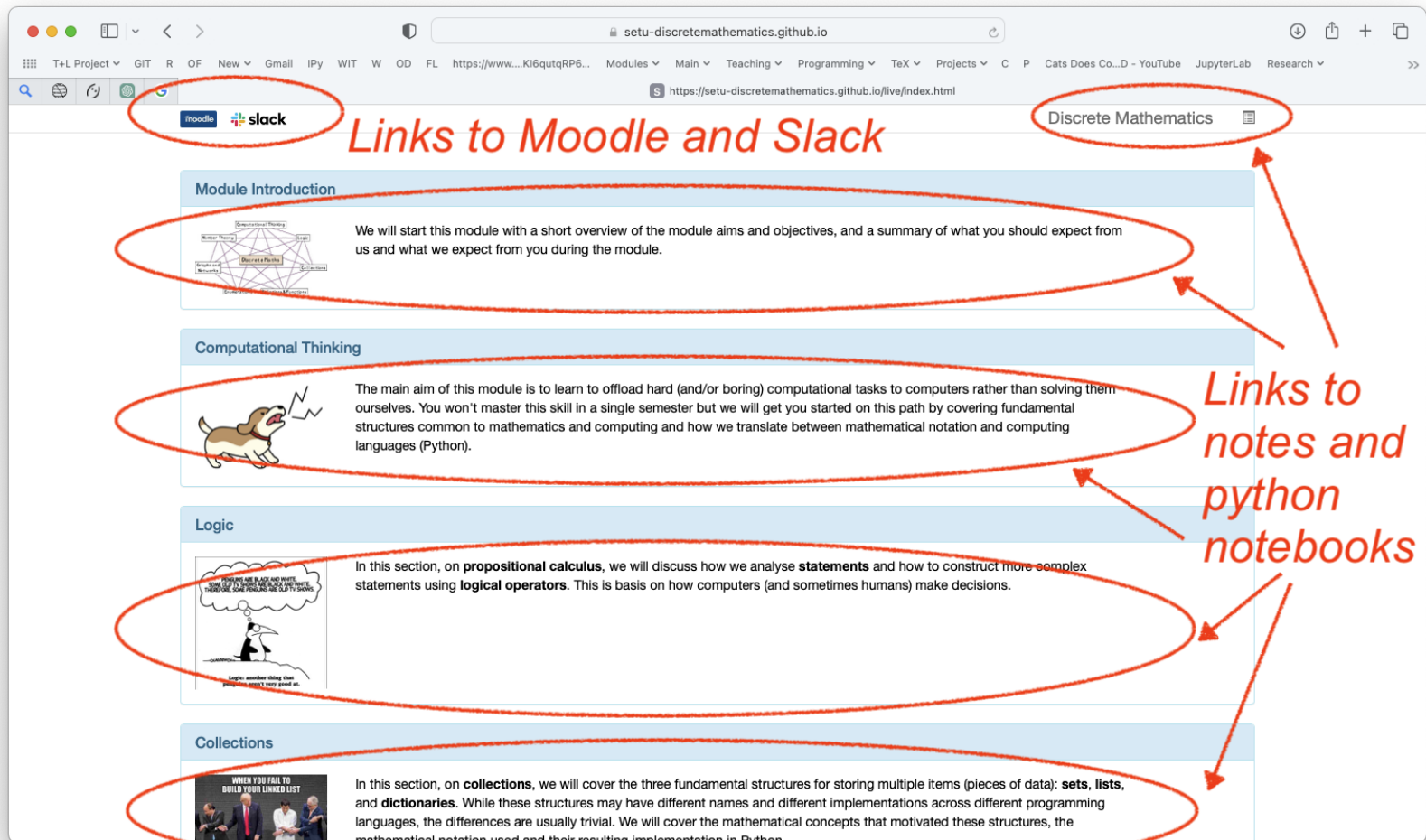
- FORUM**
  - Announcements
- Python Practicals**
  - ASSIGNMENT
  - POO - Introduction to Python and Colab

Red circles and text annotations highlight specific parts of the page:

- A red circle around the "General" section is annotated with "Links to GitHub, Slack and Colab".
- A red circle around the "Python Practicals" section is annotated with "Links to notebooks for python practicals".

# Resources — Github

- URL: [SETU-DiscreteMathematics.github.io/live](https://setu-discretemathematics.github.io/live)
- Used for all content (slides, notebooks, tutorial sheets).



The screenshot shows the website [setu-discretemathematics.github.io](https://setu-discretemathematics.github.io/live) in a browser. The address bar shows the URL. The page has a navigation bar with links to Moodle and Slack, which are circled in red. A red arrow points from the text "Links to Moodle and Slack" to these links. Another red arrow points from the text "Discrete Mathematics" to a link in the top right corner. The main content area is divided into sections: "Module Introduction", "Computational Thinking", "Logic", and "Collections". Each section has a diagram or image and a description. Red ovals are drawn around the descriptions of the "Module Introduction", "Computational Thinking", "Logic", and "Collections" sections. A red arrow points from the text "Links to notes and python notebooks" to these ovals.

**Links to Moodle and Slack**

**Discrete Mathematics**

**Links to notes and python notebooks**

**Module Introduction**

We will start this module with a short overview of the module aims and objectives, and a summary of what you should expect from us and what we expect from you during the module.

**Computational Thinking**

The main aim of this module is to learn to offload hard (and/or boring) computational tasks to computers rather than solving them ourselves. You won't master this skill in a single semester but we will get you started on this path by covering fundamental structures common to mathematics and computing and how we translate between mathematical notation and computing languages (Python).

**Logic**

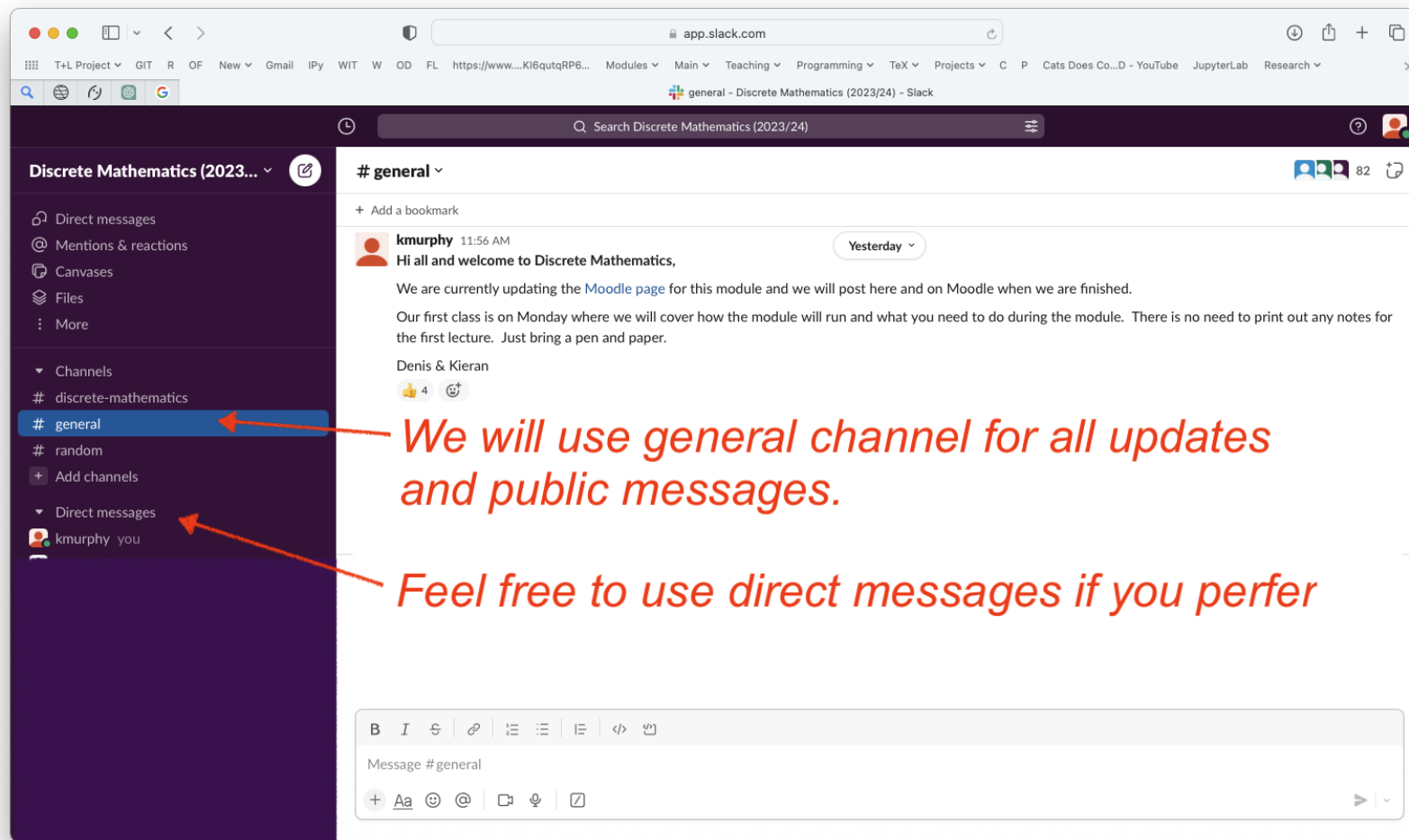
In this section, on **propositional calculus**, we will discuss how we analyse **statements** and how to construct more complex statements using **logical operators**. This is basis on how computers (and sometimes humans) make decisions.

**Collections**

In this section, on **collections**, we will cover the three fundamental structures for storing multiple items (pieces of data): **sets**, **lists**, and **dictionaries**. While these structures may have different names and different implementations across different programming languages, the differences are usually trivial. We will cover the mathematical concepts that motivated these structures, the mathematical notation used and their resulting implementation in Python.

# Resources — Slack

- URL: [discretemathe-7co3349.slack.com](https://discretemathe-7co3349.slack.com)
- Used for instant messaging, one-on-one sessions, etc.



# Resources — Colab

- We will use the online Google Colab<sup>¶</sup> environment to code in python for all of our practical work.
- You can open a notebook from these slides by clicking the "OPEN in COLAB" icon or clicking/scanning the QR code



Since using Colab is such a large part of our module we have a separate set of slides covering using Colab in more detail.

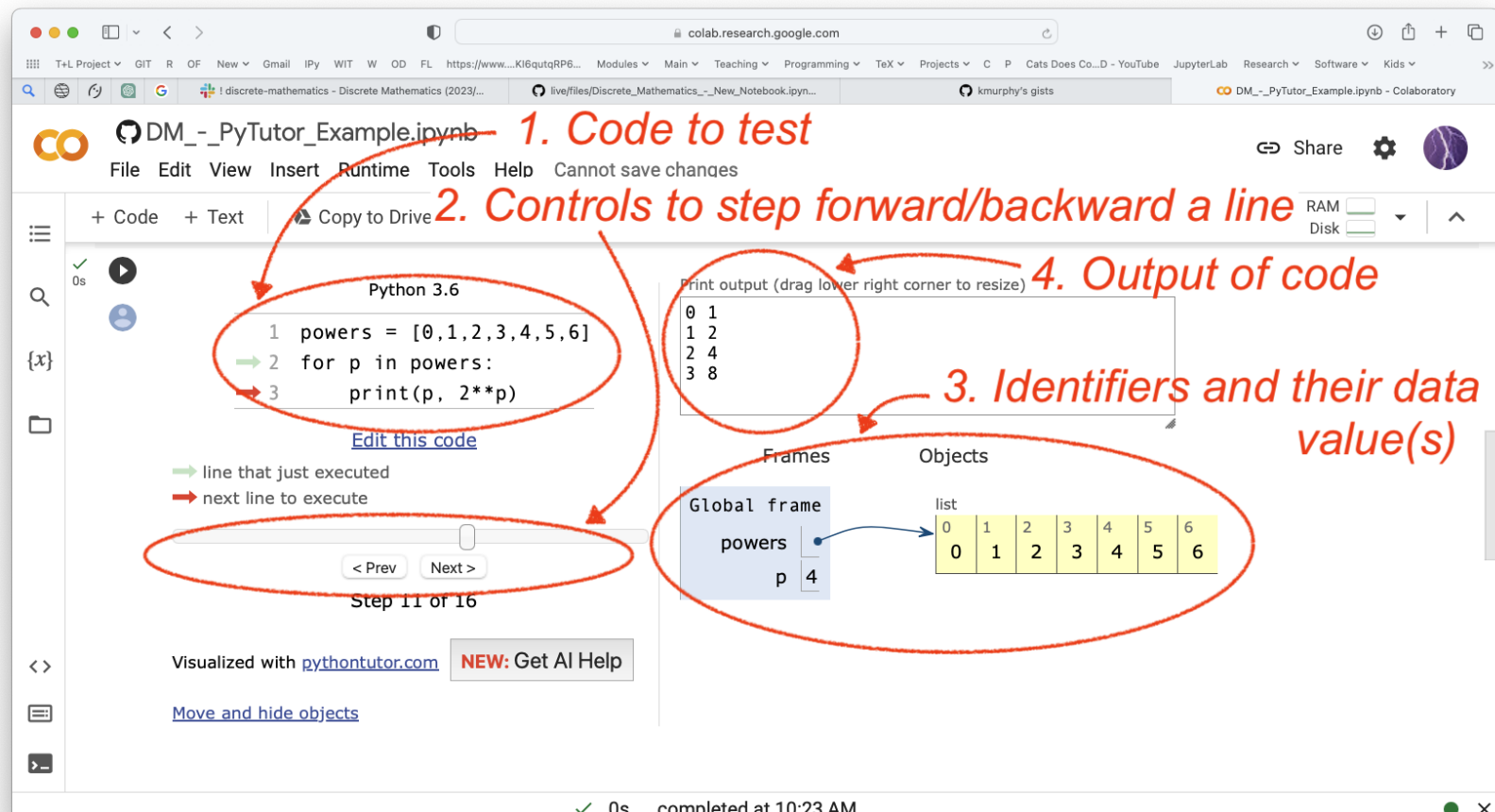


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<sup>¶</sup>Alternatively, if you want to install python on your laptop you could use the anaconda distribution from [www.anaconda.com](https://www.anaconda.com) (just install the latest 64-bit, version 3.+).

# Resources — PyTutor

- PyTutor ([pythontutor.com](https://pythontutor.com)) is a website that helps programmers to learn Python, Javascript, C/C++, and Java by visualising code execution (shows what happens to data as code runs, line by line).
- We will run PyTutor from within Colab so will demo PyTutor when we cover Colab (or click/scan the QR code).



The screenshot shows the PyTutor interface within a Google Colab notebook. The notebook is titled "DM\_-\_PyTutor\_Example.ipynb". The code being executed is:

```
1 powers = [0,1,2,3,4,5,6]
2 for p in powers:
3     print(p, 2**p)
```

Annotations in red text and circles highlight key features:

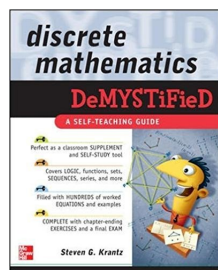
- 1. Code to test**: Points to the code editor.
- 2. Controls to step forward/backward a line**: Points to the "< Prev" and "Next >" buttons.
- 3. Identifiers and their data value(s)**: Points to the "Frames" and "Objects" panels. The "Global frame" shows a variable "powers" pointing to a list object containing [0, 1, 2, 3, 4, 5, 6].
- 4. Output of code**: Points to the "Print output" panel, which displays the results of the code execution: 0 1, 1 2, 2 4, 3 8.

Additional details visible in the interface include the "Step 11 of 16" indicator, the "Visualized with [pythontutor.com](https://pythontutor.com)" note, and the "NEW: Get AI Help" button.

# Text Books

I

I like the following textbooks on discrete mathematics and expect that my notes will overlap significantly with these books. I do encourage you to read\* them<sup>†</sup>, however, be aware they may use different notation or cover different topics.



## Discrete Mathematics Demystified

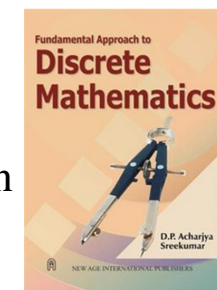
by Steven Krantz

Touches on nearly all of the topics that we hope to cover. We will probably go into greater depth in places, but a very nice and short read.

## Fundamental Approach to Discrete Mathematics

by D. P. Acharjya Sreekumar

I also liked this book, however, due to time constraints, this module only focuses on material in chapter 1–4, 8, and 10.



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\*or skim them over a coffee or two.

<sup>†</sup>I also like *Applied Discrete Structures* by Alan Doerr and Kenneth Levasseur — it is a good source of exercises. (and is free (legally))

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## Final Comments on Module

- Discrete Mathematics concepts appear either directly or indirectly in approximately 22 of the 30 modules on your degree.  
 $\implies$  *Knowing Discrete Mathematics concepts greatly simplifies rest of the course.*
- The module is intended to be an introduction to a large number of topics, so treatment is broad rather than deep.
  - ✓ Most of material is at an introductory level.
  - ⚠ Keeping in sync with material, practicals and tutorials is important.
- The continuous assessment (the practicals) is intended to reinforce the connections between programming and discrete mathematics.

The CA is a “carrot not a stick” — we want you to enjoy the module and keep up to date with the material.

